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LU-Cholesky QR algorithms for thin QR decomposition

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Abstract

This paper aims to propose the LU-Cholesky QR algorithms for thin QR decomposition (also called economy size or reduced QR decomposition). CholeskyQR is known as a fast algorithm employed for thin QR decomposition, and CholeskyQR2 aims to improve the orthogonality of a Q-factor computed by CholeskyQR. Although such Cholesky QR algorithms can efficiently be implemented in high-performance computing environments, they are not applicable for ill-conditioned matrices, as compared to the Householder QR and the Gram–Schmidt algorithms. To address this problem, we apply the concept of LU decomposition to the Cholesky QR algorithms, i.e., the idea is to use LU-factors of a given matrix as preconditioning before applying Cholesky decomposition. Moreover, we present rounding error analysis of the proposed algorithms on the orthogonality and residual of computed QR-factors. Numerical examples provided in this paper illustrate the efficiency of the proposed algorithms in parallel computing on both shared and distributed memory computers.

Keywords: Numerical linear algebra, Thin QR decomposition, High-performance computing, Rounding error analysis
2010 MSC: 65F25, 65Y05, 65G50

1. Introduction

In this paper, we propose the LU-Cholesky QR algorithms for thin QR decomposition. Suppose $A \in \mathbb{R}^{m\times n}, m \geq n$ has full column rank. The thin QR decomposition of A such that

$$A = QR, \quad Q \in \mathbb{R}^{m\times n}, \quad R \in \mathbb{R}^{n\times n}$$

is unique where Q has orthogonal columns satisfying $Q^TQ = I$ with I being the identity matrix, and R is an upper triangular matrix with positive diagonal entries. Algorithms employed for thin QR decomposition are proposed, for example Householder QR (cf. e.g. [1, p. 248]), CGS (Classical Gram-Schmidt) [2], MGS (Modified Gram-Schmidt) [2], SVQR (Singular Value QR) [3], CAQR (Communication-Avoiding QR) [4], Cholesky QR [3], and so forth.

Let u denote the unit round-off of floating-point numbers in working precision, for example, $u = 2^{-53}$ for IEEE standard 754 [5] binary64 (so-called "double precision"). Let $\kappa_2(A)$ be the generalized condition number (cf. e.g. [1, p. 284], [6]) of A such that $\kappa_2(A) := \sigma_{\max}(A)/\sigma_{\min}(A)$, where $\sigma_{\max}(A)$ and $\sigma_{\min}(A)$ are the maximum and the minimum singular values of A, respectively.

The Cholesky QR algorithms, such as CholeskyQR (cf. e.g. [1, Theorem 5.2.3]) and CholeskyQR2 [7], are ideally employed for thin QR decomposition due to their communication avoidance for tall-skinny matrices. CholeskyQR2 first applies CholeskyQR to A, and then applies it again to the computed Q-factor in order to refine the orthogonality. A rounding error analysis of CholeskyQR2 is presented in [8]. Computational kernels of Householder QR, CGS, MGS, and CAQR algorithms are BLAS-level 1 and -level 2 routines. On the other hand, the Cholesky QR algorithms can be implemented mainly using standard numerical linear algebra libraries, such as BLAS-level 3 and LAPACK routines, which entails for their high computational performance and the parallelization efficiency. The major drawback of the Cholesky QR algorithms is to square the generalized condition number of a given problem, i.e., $\kappa_2(A^TA) = \kappa_2(A)^2$, since they directly compute a Cholesky decomposition of the matrix A^TA. Let $\hat{B}$ be a computed result of the matrix multiplication A^TA in floating-point arithmetic. If $\kappa_2(A) > \sqrt{u^{-1}}$, then $\kappa_2(\hat{B}) > u^{-1}$ is expected. In this case, Cholesky decomposition of $\hat{B}$ tends to fail, so that the Cholesky QR algorithms are not applicable.

To solve the problem, we propose two algorithms for thin QR decomposition using LU decomposition, herein referred to LU-CholeskyQR and LU-CholeskyQR2. Our focus is on Doolittle's LU decomposition of a matrix A such that $PA = LU$, where $L \in \mathbb{R}^{m\times n}$ is a unit lower triangular matrix, $U \in \mathbb{R}^{n\times n}$ is an upper triangular matrix, and $P \in \mathbb{R}^{m\times m}$ is a permutation matrix. After the LU decomposition of A, Cholesky decomposition is used for the matrix L^TL. Then, it is likely that it runs to completion, since L tends to be fairly well-conditioned even if A is

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ill-conditioned (cf. e.g. [1, p. 142], [9, p. 297]). This suggests that we can apply CholeskyQR to L even if $\kappa_2(A) > \sqrt{u^{-1}}$ is satisfied. We call this algorithm by LU-CholeskyQR. We also develop an algorithm called LU-CholeskyQR2, which is a refinement of LU-CholeskyQR in terms of the orthogonality of a Q-factor computed by LU-CholeskyQR.

The rest of the paper is organized as follows. In Section 2, we introduce the notation used in this paper and mention the previous studies. In Section 3, we propose the LU-Cholesky QR algorithms together with results of their rounding error analysis. In Section 4, numerical examples are presented to illustrate the efficiency of the proposed algorithms in terms of the orthogonality of computed Q-factors, residuals of computed QR-factors, and the computing times on both shared and distributed memory computers. Finally, we conclude the paper in Section 5. Besides, we present rounding error analysis of the proposed algorithms on the orthogonality and residual in Appendix A.

2. Background of the study

2.1. Notation

We introduce the notation used in this paper. Let $\mathbb{F}$ be a set of floating-point numbers as defined by the IEEE Std. 754 [5]. The notation $fl(\cdot)$ indicates that all operations inside the parentheses are evaluated using floating-point arithmetic with the round-to-nearest (ties to even) mode. We count the number of floating-point operations by flops[1]. For simplicity, only the expression with the maximum degree is written for a polynomial for flops. For example, the cost for matrix multiplication is simply noted by $2n^3$ flops instead of $2n^3 - n^2$ flops.

2.2. Cholesky QR algorithms

CholeskyQR is known as a fast algorithm for the thin QR decomposition of full column rank matrices. For a full column rank matrix $A \in \mathbb{R}^{m\times n}$, $m \geq n$, if $B = A^TA = R^TR$, where $R \in \mathbb{R}^{n\times n}$ is the Cholesky-factor of A^TA, then $Q = AR^{-1} \in \mathbb{R}^{m\times n}$ defines the thin QR decomposition of A such that $A = QR$ [1, Theorem 5.2.3]. A thin QR decomposition of A in floating-point arithmetic aims to compute such QR-factors as $A \approx \hat{Q}\hat{R}$, where $\hat{Q} \in \mathbb{F}^{m\times n}$ has approximately orthogonal columns and $\hat{R} \in \mathbb{F}^{n\times n}$ is an upper triangular matrix.

Here, CholeskyQR is introduced in MATLAB-like notations.

Algorithm 1. *CholeskyQR (cf. e.g. [1, Theorem 5.2.3])*
For a full column rank matrix $A \in \mathbb{F}^{m\times n}$, the following algorithm produces computed thin QR-factors such that $A \approx \hat{Q}_1\hat{R}_1$.

function $[\hat{Q}_1, \hat{R}_1] = \mathrm{CholQR}(A)$
  $\hat{B} = A' * A$; $\quad \% \ \hat{B} = fl(A^TA)$
  $\hat{R}_1 = \mathrm{chol}(\hat{B})$; $\quad \%$ *Cholesky decomposition* $\hat{B} \approx \hat{R}_1^T\hat{R}_1$
  $\hat{Q}_1 = A/\hat{R}_1$; $\quad \%$ *Solve* $Q_1\hat{R}_1 = A$ *for* Q_1
end

[1] It is not FLOPS, Floating-point Operation Per Second.

Here, $\mathrm{chol}(\hat{B})$ produces an upper triangular matrix as a computed Cholesky-factor of $\hat{B}$. The computational cost for $\hat{B} = A' * A$ in Algorithm 1 computed by `dsyrk` in BLAS is mn^2 flops. Moreover, the cost for $\mathrm{chol}(\hat{B})$ and $A/\hat{R}_1$ are $n^3/3$ and mn^2 flops, respectively. Hence, the total cost for Algorithm 1 is $2mn^2 + n^3/3$ flops. The matrix $\hat{B}$ becomes ill-conditioned if $\kappa_2(A) > \sqrt{u^{-1}}$, and it is likely that Cholesky decomposition of $\hat{B}$ breaks down as it gives a square root of a negative number.

Subsequently, CholeskyQR2 refines the orthogonality of a computed Q-factor $\hat{Q}_1$ obtained by CholeskyQR.

Algorithm 2. *CholeskyQR2 [7]*
For a full column rank matrix $A \in \mathbb{F}^{m\times n}$, the following algorithm produces computed thin QR-factors such that $A \approx \hat{Q}_2\hat{R}_2$.

function $[\hat{Q}_2, \hat{R}_2] = \mathrm{CholQR2}(A)$
  $[\hat{Q}_1, \hat{R}_1] = \mathrm{CholQR}(A)$; $\quad \% \ A \approx \hat{Q}_1\hat{R}_1$
  $[\hat{Q}_2, \tilde{R}] = \mathrm{CholQR}(\hat{Q}_1)$; $\quad \% \ \hat{Q}_1 \approx \hat{Q}_2\tilde{R}$
  $\hat{R}_2 = \tilde{R} * \hat{R}_1$;
end

The computational cost for Algorithm 2 is $4mn^2 + n^3$ flops. This algorithm provides computed QR-factors $\hat{Q}_2, \hat{R}_2$ such that $A \approx \hat{Q}_1\hat{R}_1 \approx \hat{Q}_2\tilde{R}\hat{R}_1 \approx \hat{Q}_2\hat{R}_2$. A rounding error analysis of Algorithms 1 and 2 is given in [8]. Here, assuming that the condition

$$\delta := 8\kappa_2(A)\sqrt{(mn + n(n+1))u} \leq 1, \tag{1}$$

is satisfied, it holds that

$$\|\hat{Q}_1^T\hat{Q}_1 - I\|_2 \leq \frac{5}{64}\delta^2, \tag{2}$$

$$\|\hat{Q}_2^T\hat{Q}_2 - I\|_2 \leq 6(mn + n(n+1))u, \tag{3}$$

provided that neither underflow nor overflow occurs in floating-point computations. The orthogonality of $\hat{Q}_2$ is thus refined as compared to $\hat{Q}_1$. As can be seen, the orthogonality of $\hat{Q}_1$ in (2) depends on $\kappa_2(A)$, while that of $\hat{Q}_2$ does not, i.e., $\|\hat{Q}_2^T\hat{Q}_2 - I\|_2$ is bounded by $O(u)$ regardless of $\kappa_2(A)$ under the condition (1). However, as can also be seen from (1), the above Cholesky QR algorithms cannot be applied if $\kappa_2(A) > \sqrt{u^{-1}}$.

3. Proposed algorithms

Suppose $A \in \mathbb{F}^{m\times n}$ has full column rank. Our idea is to avoid Cholesky decomposition of A^TA. In fact, it is possible using Doolittle's LU decomposition of A such that $PA = LU$, where L is a unit lower triangular matrix, U is an upper triangular matrix, and P is a permutation matrix. It is heuristically expected that L is fairly well-conditioned, even if A is ill-conditioned (cf. e.g. [1, p. 142], [9, p. 297]). Then, if we obtain QR-factors of P^TL such that $P^TL = QS$ where Q is orthogonal and S is upper triangular, then $A = P^TLU = QSU =: QR$. Here, QR-factors of P^TL can efficiently be obtained by CholeskyQR through Cholesky decomposition of $L^TPP^TL = L^TL$. It turns

out that Doolittle's LU decomposition can work as a preconditioning for the Cholesky QR algorithms.

Let $\hat{L}$, $\hat{U}$, and P be computed LU-factors of A such that $PA \approx \hat{L}\hat{U}$. Then, we perform Cholesky decomposition of $fl(\hat{L}^T\hat{L})$ in floating-point arithmetic, which is expected to run to completion. With this, our algorithms can produce computed QR-factors of A even if $\kappa_2(A) > \sqrt{u^{-1}}$, which means our algorithms are applicable to a wider problem class than the original Cholesky QR algorithms.

The following algorithm called LU-CholeskyQR provides computed QR-factors utilizing LU-factors.

Algorithm 3. *LU-CholeskyQR*
For a full column rank matrix $A \in \mathbb{F}^{m\times n}$, the following algorithm produces computed thin QR-factors such that $A \approx \hat{Q}_1\hat{R}_1$.

function $[\hat{Q}_1, \hat{R}_1] = \text{LU_CholQR}(A)$
  $[\hat{L}, \hat{U}, P] = \text{lu}(A)$; % $PA \approx \hat{L}\hat{U}$ (P is not used hereafter.)
  $\hat{B} = L' * L$; % $\hat{B} = fl(\hat{L}^T\hat{L})$
  $\hat{S} = \text{chol}(\hat{B})$; % Cholesky decomposition $\hat{B} \approx \hat{S}^T\hat{S}$
  $\hat{R}_1 = \hat{S} * \hat{U}$;
  $\hat{Q}_1 = A/\hat{R}_1$; % Solve $\hat{Q}_1\hat{R}_1 = A$ for $\hat{Q}_1$
end

Since the computational cost for LU decomposition of A is $2mn^2 - n^3/3$ flops, the total cost for Algorithm 3 is $4mn^2 - n^3/3$ flops.

Next, we propose a variant of LU-CholeskyQR algorithm.

Algorithm 4. *A variant of LU-CholeskyQR*
For a full column rank matrix $A \in \mathbb{F}^{m\times n}$, the following algorithm produces computed thin QR-factors such that $A \approx \hat{Q}_1\hat{R}_1$.

function $[\hat{Q}_1, \hat{R}_1] = \text{variant_LU_CholQR}(A)$
  $[\hat{L}, \hat{U}, P] = \text{lu}(A)$; % $PA \approx \hat{L}\hat{U}$
  $\hat{B} = L' * L$; % $\hat{B} = fl(\hat{L}^T\hat{L})$
  $\hat{S} = \text{chol}(\hat{B})$; % Cholesky decomposition $\hat{B} \approx \hat{S}^T\hat{S}$
  $\hat{Q}_1 = (P' * \hat{L})/\hat{S}$; % P is the permutation matrix
  $\hat{R}_1 = \hat{S} * \hat{U}$;
end

Remark that Algorithm 1 is applicable for $\hat{L}$ such that

$$A \approx P^T\hat{L}\hat{U} \approx P^T\hat{Q}\hat{R}\hat{U},$$

where $P^T\hat{Q} =: \tilde{Q}_1$ has approximately orthogonal columns, $\hat{R}\hat{U} =: \tilde{R}_1$ is an upper triangular matrix, and $\hat{Q}, \hat{R}$ are QR-factors computed by Algorithm 1. This achieves numerical stability, even if A is ill-conditioned. In addition, an upper bound of the orthogonality of $\hat{Q}_1$ is given as

$$\|\hat{Q}_1^T\hat{Q}_1 - I\|_2 \le \kappa_2(\hat{L})^2(mn + n(n+1))u, \tag{4}$$

from (2), if $8\kappa_2(\hat{L})\sqrt{(mn + n(n+1))u} \le 1$. However, numerical stability of Algorithm 4 is the same as that of Algorithm 3, since numerical stability depends on applicability of Cholesky decomposition for $\hat{L}^T\hat{L}$. The residual of the QR-factors computed by Algorithm 4 is worse than that of Algorithm 3, which is stated in subsequent Theorems 4. Upper bounds for residuals of QR-factors computed by Algorithms 3 and 4 depend on $\|A\|_2$ and $\|L\|_2\|U\|_2$, respectively. Hence, QR-factors computed by Algorithm 3 has a better upper bound of residual than that computed by Algorithm 4. Therefore, Algorithm 3 is suitable for preconditioning in many cases.

Next, the following algorithm called LU-CholeskyQR2 refines the orthogonality computed by $\hat{Q}_1$ obtained by Algorithm 3.

Algorithm 5. *LU-CholeskyQR2*
For a full column rank matrix $A \in \mathbb{F}^{m\times n}$, the following algorithm produces computed thin QR-factors such that $A \approx \hat{Q}_2\hat{R}_2$ using Algorithm 3.

function $[\hat{Q}_2, \hat{R}_2] = \text{LU_CholQR2}(A)$
  $[\hat{Q}_1, \hat{R}_1] = \text{LU_CholQR}(A)$; % $A \approx \hat{Q}_1\hat{R}_1$ (Algorithm 3)
  $[\hat{Q}_2, \tilde{R}] = \text{CholQR}(\hat{Q}_1)$; % $\hat{Q}_1 \approx \hat{Q}_2\tilde{R}$
  $\hat{R}_2 = \tilde{R} * \hat{R}_1$;
end

Thus, the orthogonality of $\hat{Q}_1$ is refined by CholeskyQR as $\hat{Q}_1 \approx \hat{Q}_2\tilde{R}$. The computational cost for Algorithm 5 is $6mn^2 + n^3/3$ flops.

In a similar way to the Cholesky QR algorithms, our LU-Cholesky QR algorithms (Algorithms 3 and 5) can be implemented with standard numerical linear algebra libraries, such as BLAS and LAPACK on shared memory computers and PBLAS and ScaLAPACK on distributed memory computers. Therefore, our proposed algorithms effectively benefit from highly optimized routines in such libraries, in particular, in parallel computing. However, while Cholesky QR has large grain parallelism, LU-Cholesky QR does not, because it uses LU decomposition with pivoting. Accordingly, in a parallel environment with large communication latency, LU-Cholesky QR can be much slower than Cholesky QR.

As results of rounding error analysis of Algorithms 3 and 5 on the orthogonality and the residual of computed QR-factors, we present the following theorems and corollary.

Theorem 1. *Let $\hat{Q}_1$ be obtained by Algorithm 3, where neither overflow nor underflow occurs in floating-point operations. Under the assumptions*

$$\delta_L := 8\kappa_2(\hat{L})\sqrt{(mn + n(n+1))u} \le 1 \tag{5}$$

$$\delta_{LU} := 64\kappa_2(\hat{L})\kappa_2(\hat{U})n^2u \le 1, \tag{6}$$

it holds

$$\|\hat{Q}_1^T\hat{Q}_1 - I\|_2 \le \frac{1}{8}\max(\delta_{LU}, \delta_L^2). \tag{7}$$

Corollary 2. *Let $\hat{Q}_2$ be obtained by Algorithm 5, where neither overflow nor underflow occurs in floating-point operations. Under the assumptions* (5)*,* (6)*, and*

$$8\kappa_2(\hat{Q}_1)\sqrt{(mn + n(n+1))u} \le 1,$$

then

$$\|\hat{Q}_2^T \hat{Q}_2 - I\|_2 \leq 6.5(mn + n(n+1))u.$$

Theorem 3. *Let $\hat{Q}_2, \hat{R}_2$ be obtained by Algorithm 5, where neither overflow nor underflow occurs in floating-point operations. Under the assumptions* (5) *and* (6)*,*

$$\|\hat{Q}_2\hat{R}_2 - A\|_2 \leq 4.09n^2u\|A\|_2. \tag{8}$$

Theorem 4. *Let $\hat{Q}_1, \hat{R}_1$ be obtained by Algorithm 4, where neither overflow nor underflow occurs in floating-point operations. Under the assumptions* (5) *and* (6)*,*

$$\|\hat{Q}_1\hat{R}_1 - A\|_2 \leq 3.15n^2u\|\hat{L}\|_2\|\hat{U}\|_2. \tag{9}$$

We will provide proofs of Theorems 1, 3, and 4 and Corollary 2 in Appendix A. From Theorem 4, the residual norm of QR-factors of Algorithm 4 becomes worse, that depends on $\|\hat{L}\|_2\|\hat{U}\|_2 (\gtrsim \|A\|_2)$. Hence, for residual norm, Algorithm 3 is better than Algorithm 4, and orthogonality can be refined as Algorithm 5. Then, we adapt Algorithm 3 hereafter.

4. Numerical results

We present some numerical results of our proposed algorithms in both shared and distributed memory computer environments.

4.1. Shared memory computer environments

We compare the orthogonality of computed Q-factors, residual norms of computed QR-factors, and computing times for the above algorithms (Algorithms 1, 2, 3, and 5) and a standard Householder QR algorithm through numerical examples in the following two shared memory computer environments:

Env. 1 CPU: Intel(R) Core(TM) i7-8550U, 4 cores, Memory: 16 GB, OS: Windows 10, Software: MATLAB R2018a

Env. 2 CPU: Intel(R) Core(TM) i9-7900X, 10 cores, Memory: 128 GB, OS: Windows 10, Software: MATLAB R2018a

We first compare computing time, and a test matrix $A \in \mathbb{F}^{m\times n}$ is generated using MATLAB's function as `A = rand(m,n)`. With respect to the computing times, it is apparent from Figs. 1 and 2 that the proposed algorithms are fairly faster than MATLAB's `qr` function. Moreover, when n gets to a certain size, the cost for preconditioning by LU decomposition is relatively less expensive, because the computational performance of LU decomposition is improved for larger n, especially in Env. 2.

Next, we compare strong-scalability of MATLAB's `qr` function, CholeskyQR2, and LU-CholeskyQR2 algorithms on Env. 2. From Fig. 3, strong-scalability of LU-CholeskyQR2 is better than that of MATLAB's QR, and comparable to that of CholeskyQR2.

Next we compare orthogonality and residual on Env. 1, and generate test matrices using Higham's `randsvd` function [9] as

```
A = gallery('randsvd',[m,n],cnd,mode,m,n,1),
```

where `cnd` is a speficied generalized condition number $\kappa_2(A)$, and `mode` can be chosen as follows:

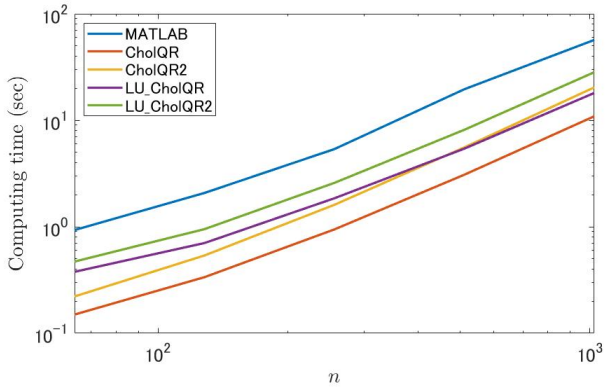


Figure 1: Comparison of the computing times for various n with $m = 500{,}000$ in Env. 1

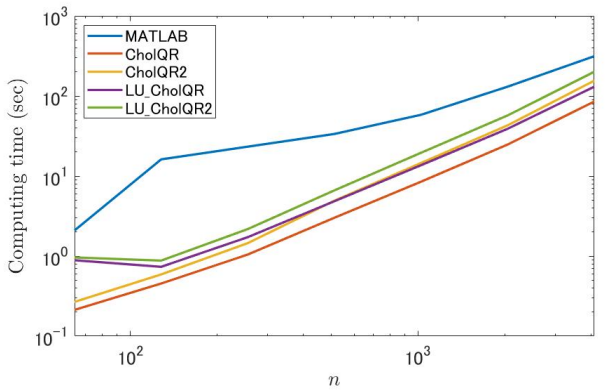


Figure 2: Comparison of the computing times for various n with $m = 1{,}000{,}000$ in Env. 2

1. One large singular value.
2. One small singular value.
3. Geometrically distributed singular values.
4. Arithmetically distributed singular values.
5. Random singular values with uniformly distributed logarithm.

In numerical examples, we choose `mode = 3` unless otherwise specified. As a standard Householder QR algorithm, we use MATLAB's `qr` function for thin QR decomposition as

```
[Q,R] = qr(A,0).
```

For comparison on the orthogonality $\|\hat{Q}^T\hat{Q} - I\|_2$ and the residual norms $\|\hat{Q}\hat{R} - A\|_2$ for computed QR-factors, we use the Advanpix Multiprecision Computing Toolbox [10] for calculating them precisely.

Figure 4 compares the orthogonality as $\|\hat{Q}^T\hat{Q} - I\|_2$, and Fig. 5 compares the residuals as $\|\hat{Q}\hat{R} - A\|_2$. As can be seen, the proposed algorithms (LU-CholeskyQR and LU-CholeskyQR2) run to completion even if $\kappa_2(A) > \sqrt{u^{-1}}$. Moreover, the Q-factors computed by LU-CholeskyQR are successfully refined by LU-CholeskyQR2 in terms of the orthogonality. Note that

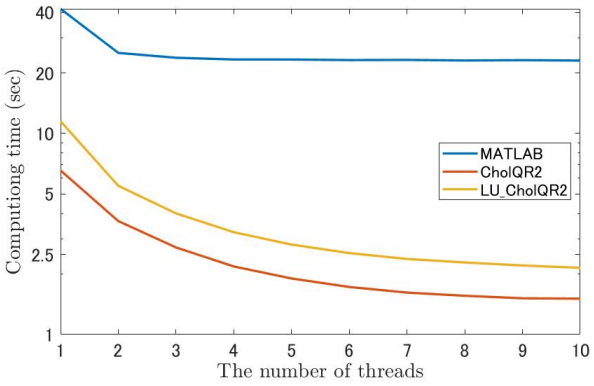


Figure 3: Comparison of the computing times for various threads with m = 1,000,000, n = 256 in Env. 2

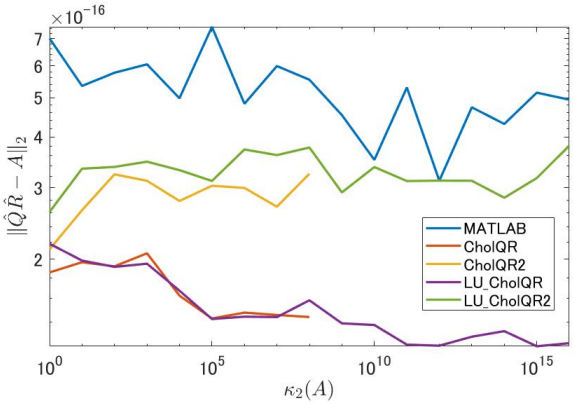


Figure 5: Comparison of the residual norms $\|\hat{Q}\hat{R} - A\|_2$ for various $\kappa_2(A)$ with m = 1024, n = 128 in Env. 1

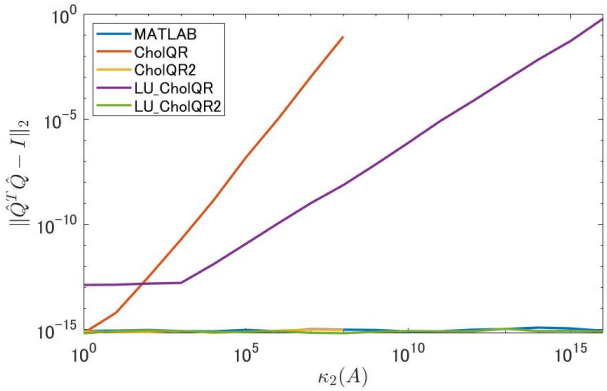


Figure 4: Comparison of the orthogonality $\|\hat{Q}^T\hat{Q} - I\|_2$ for various $\kappa_2(A)$ with m = 1024, n = 128 in Env. 1

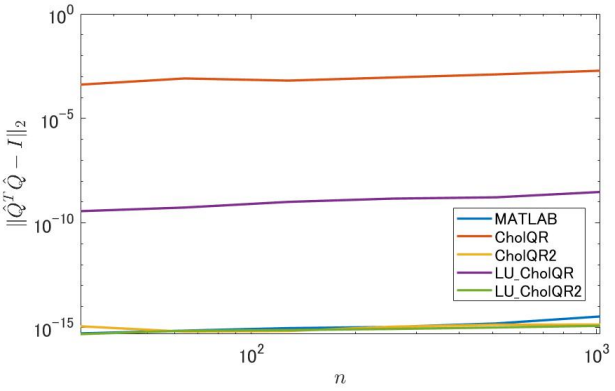


Figure 6: Comparison of the orthogonality $\|\hat{Q}^T\hat{Q} - I\|_2$ for various n with m = 1024, cnd = 10^7 in Env. 1

neither CholeskyQR nor CholeskyQR2 cannot produce QR-factors if $\kappa_2(A) > \sqrt{u^{-1}}$, as Cholesky decomposition of $fl(A^TA)$ breaks down. Both the orthogonality and the residual norms of computed QR-factors obtained by LU-CholeskyQR2 are comparable to the results produced by MATLAB's `qr` function.

Moreover, Figs. 6 and 7 compare the orthogonality and the residual norms for $m = 1024$, $32 \le n \le 1024$, and $\kappa_2(A) \approx 10^7$. Similar to the previous results, both the orthogonality and the residual norms of the computed QR-factors obtained by LU-CholeskyQR2 are comparable to the results produced by MATLAB's `qr` function.

Furthermore, Figs. 8 and 9 display the orthogonality and the residual norms of the QR-factors computed by LU-CholeskyQR2, respectively, for all `mode` $\in \{1, 2, 3, 4, 5\}$. When $\kappa_2(A) \approx u^{-1}$, the orthogonality becomes slightly worse in the cases of mode $\in \{1, 2\}$. However, the residual norms are still small for all the modes.

4.2. *Distributed memory computer environments*

Finally, we show some numerical results on parallel distributed memory computers using RIKEN's K computer and FUJITSU Supercomputer PRIMEHPC FX100. Computing times of the above algorithms are compared in the following two computational environments:

K computer CPU: SPARC6^{TM} VIIIfx (8 cores), Memory: 16 GB / node

FX100 CPU: SPARC6^{TM} XIfx (32 cores), Memory: 32 GB (HMC) / node

Here, we use `mpifccpx` as the Fujitsu C compiler command with MPI with the options

```
-Kfast,parallel,openmp -SCALAPACK -SSL2BLAMP
```

on K computer and FX100 in common.

We generate $m \times n$ matrices whose elements are pseudo-random numbers uniformly distributed in the interval $(0, 1)$ with m = 1,048,576, n = 256. Note that the generated test matrices are not ill-conditioned, so that the Cholesky QR algorithms are applicable.

Tables 1 and 2 compare the computing times for the Householder QR algorithm (labeled 'HouseholderQR'), CholeskyQR2, and LU-CholeskyQR2. We use the ScaLAPACK routines listed below:

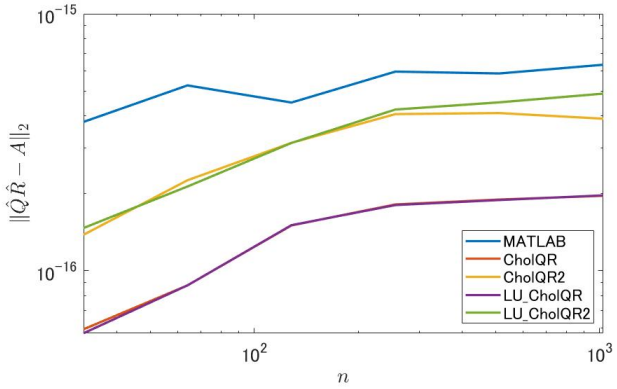


Figure 7: Comparison of the residual norms $\|\hat{Q}\hat{R} - A\|_2$ for various n with $m = 1024$, cnd $= 10^7$ in Env. 1

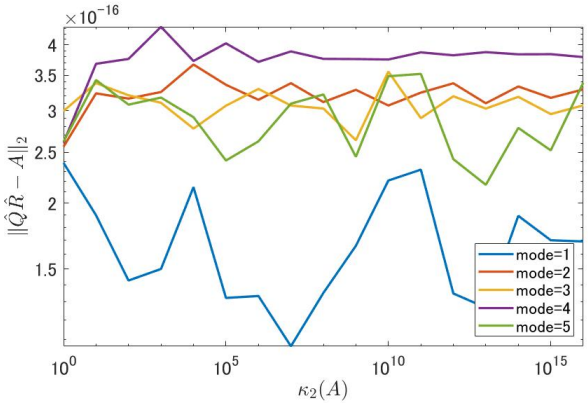


Figure 9: The residual norms $\|\hat{Q}\hat{R}-A\|_2$ for Algorithm 5 (LU-CholeskyQR2) for various $\kappa_2(A)$ with several singular value distributions with $m = 1024$, $n = 128$

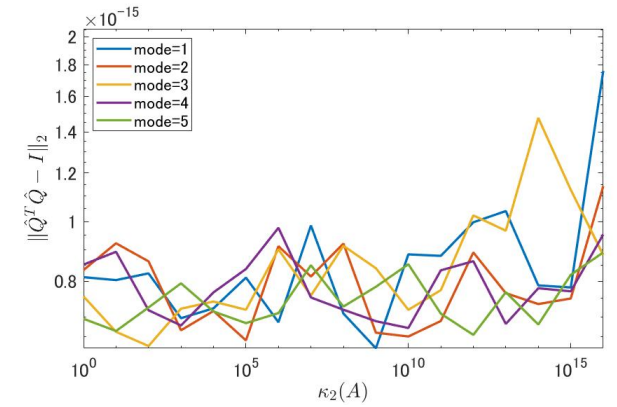


Figure 8: The orthogonality $\|\hat{Q}^T\hat{Q}-I\|_2$ for Algorithm 5 (LU-CholeskyQR2) for various $\kappa_2(A)$ with several singular value distributions with $m = 1024$, $n = 128$

Algorithm	Major ScaLAPACK routines
HouseholderQR	`pdgeqrf`, `pdorgqr`
CholeskyQR2	`dsyrk`, `dpotrf`, `dtrsv`, `dtrmm` $(*)$
LU-CholeskyQR2	`pdgetrf` and all routines in $(*)$

As can be seen, CholeskyQR2 is the fastest among them in all the cases, as expected. Although the computational performance of Doolittle's LU decomposition is not very high due to partial pivoting, the proposed algorithm (LU-CholeskyQR2) is faster than the Householder QR algorithm (HouseholderQR) in almost all the cases.

5. Conclusion

We proposed the LU-Cholesky QR algorithms for thin QR decomposition to improve the existing CholeskyQR algorithms in terms of their robustness. On the basis of our investigations in this paper, the proposed algorithms would effectively work for ill-conditioned matrices where the Cholesky QR algorithms, such as CholeskyQR and CholeskyQR2, are not applicable.

With respect to computing time, the comparisons in our numerical examples revealed that

- LU-CholeskyQR2 is faster than the Householder QR algorithm on both our shared memory computers and the distributed memory computers used in this paper,
- the computing time for LU-CholeskyQR2 is about 1.5 times more than that of CholeskyQR2 on our shared memory computers, and
- the computing time for LU-CholeskyQR2 is between 3 and 5 times more than that of CholeskyQR2 on the distributed memory computers used in this paper.

With respect to the orthogonality and norms of residuals, LU-CholeskyQR2 is comparable to CholeskyQR2 and the Householder QR algorithm.

In addition, we presented the results of rounding error analysis of the proposed algorithms in a similar way to the Cholesky QR algorithms.

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Appendix A. Rounding error analysis of the proposed algorithms

We present rounding error analysis of Algorithms 3 and 5 on the orthogonality of computed Q-factors, i.e., we provide the proofs of Theorem 1 and Corollary 2.

Table 1: Comparison of computing times (sec) on RIKEN's K computer (m = 1,048,576, n = 256)

Algorithm \ # nodes	1	2	4	8	16	32	64	128
HouseholderQR	33.89	27.01	12.51	6.10	3.09	1.64	0.89	0.86
CholeskyQR2	7.78	2.58	1.29	0.67	0.36	0.20	0.14	0.12
LU-CholeskyQR2	23.93	12.39	6.24	2.79	1.48	0.82	0.50	0.43

Table 2: Comparison of computing times (sec) on FUJITSU FX100 (m = 1,048,576, n = 256)

Algorithm \ # nodes	1	2	4	8	16	32	64
HouseholderQR	15.75	8.12	3.97	2.03	1.09	0.62	0.35
CholeskyQR2	1.79	0.94	0.50	0.27	0.16	0.11	0.13
LU-CholeskyQR2	7.97	3.36	1.74	0.68	0.57	0.42	0.38

For $X = (x_{ij}), Y = (y_{ij}) \in \mathbb{R}^{m\times n}$, the notation $|X|$ means $|X| = (|x_{ij}|) \in \mathbb{R}^{m\times n}$. The inequality $X \le Y$ means $x_{ij} \le y_{ij}$ for all (i, j).

We assume that all floating-point operations are done with unit round-off u, neither overflow nor underflow occur in the floating-point operations, and any divide-and-conquer method is not used for matrix multiplication.

Appendix A.1. Preliminaries

We begin with standard rounding error analysis for matrix multiplication, LU decomposition, and triangular systems. This covers in particular blocking strategies, but not more sophisticated methods, such as Strassen [11], Coppersmith and Winograd [12], or Williams [13]. For example, given $A \in \mathbb{F}^{m\times n}$ and $B \in \mathbb{F}^{n\times k}$, the product AB is computed by using mk inner products in dimension n in any order of evaluation.

Lemma 5 (Jeannerod–Rump [14]). *For matrices $A \in \mathbb{F}^{m\times n}$ and $B \in \mathbb{F}^{n\times k}$, a computed result of matrix multiplication $C := fl(AB) \in \mathbb{F}^{m\times k}$ satisfies*

$$|AB - C| \le nu|A||B|.$$

Lemma 6 (Rump–Jeannerod [15]). *Suppose that $\hat{L} \in \mathbb{F}^{m\times n}$ and $\hat{U} \in \mathbb{F}^{n\times n}$ are computed LU-factors of $A \in \mathbb{F}^{m\times n}$. Then,*

$$\hat{L}\hat{U} - A = E_5, \quad |E_5| \le nu|\hat{L}||\hat{U}|.$$

Lemma 7 (Rump–Jeannerod [15]). *Suppose that $\hat{R} \in \mathbb{F}^{n\times n}$ is a computed Cholesky-factor of $A \in \mathbb{F}^{n\times n}$. Then,*

$$\hat{R}^T\hat{R} - A = E_5, \quad |E_5| \le (n+1)u|\hat{R}^T||\hat{R}|.$$

Lemma 8 (Rump–Jeannerod [15]). *For a nonsingular triangular matrix $T \in \mathbb{F}^{n\times n}$ and $B \in \mathbb{F}^{n\times k}$, suppose triangular systems $TX = B$ are solved by forward or backward substitution. Then, a computed solution $\hat{X} \in \mathbb{F}^{n\times k}$ satisfies*

$$T\hat{X} - B = \Delta, \quad |\Delta| \le nu|T||\hat{X}|.$$

For $A \in \mathbb{F}^{m\times n}$, $m \ge n$, suppose that matrices $\hat{L} \in \mathbb{F}^{m\times n}$, $\hat{U} \in \mathbb{F}^{n\times n}$, and $P \in \mathbb{F}^{m\times m}$ are computed by Doolittle's LU decomposition with partial pivoting such that $\hat{L}\hat{U} \approx PA$. From Lemmas 5, 6, 7, and 8,

$$PA - \hat{L}\hat{U} = E_1, \quad |E_1| \le nu|\hat{L}||\hat{U}|, \tag{A.1}$$
$$\hat{B} - \hat{L}^T\hat{L} = E_2, \quad |E_2| \le mu|\hat{L}^T||\hat{L}|, \tag{A.2}$$
$$\hat{S}^T\hat{S} - \hat{B} = E_3, \quad |E_3| \le (n+1)u|\hat{S}^T||\hat{S}|, \tag{A.3}$$
$$\hat{R}_1 - \hat{S}\,\hat{U} = E_4, \quad |E_4| \le nu|\hat{S}\,||\hat{U}|, \tag{A.4}$$
$$\hat{Q}_1\hat{R}_1 - A = E_5, \quad |E_5| \le nu|\hat{Q}_1||\hat{R}_1|, \tag{A.5}$$

are satisfied for the matrices in Algorithm 3. It is known [9, p. 111] that, for $M, N \in \mathbb{R}^{m\times n}$ with $|M| \le N$,

$$\|M\|_2 \le \|\,|M|\,\|_2 \le \sqrt{\mathrm{rank}(M)}\|M\|_2, \tag{A.6}$$
$$\|M\|_2 \le \|N\|_2. \tag{A.7}$$

Therefore, $E_1, \ldots, E_5$ satisfy the following inequalities.

$$\|E_1\|_2 \le n^2u\|\hat{L}\|_2\|\hat{U}\|_2, \tag{A.8}$$
$$\|E_2\|_2 \le mnu\|\hat{L}\|_2^2, \tag{A.9}$$
$$\|E_3\|_2 \le n(n+1)u\|\hat{S}\,\|_2^2, \tag{A.10}$$
$$\|E_4\|_2 \le n^2u\|\hat{S}\,\|_2\|\hat{U}\|_2, \tag{A.11}$$
$$\|E_5\|_2 \le n^2u\|\hat{Q}_1\|_2\|\hat{R}_1\|_2. \tag{A.12}$$

Assume that

$$\delta_L := 8\kappa_2(\hat{L})\sqrt{(mn + n(n+1))u} \le 1, \tag{A.13}$$
$$\delta_{LU} := 64\kappa_2(\hat{L})\kappa_2(\hat{U})n^2u \le 1 \tag{A.14}$$

are satisfied. From the assumptions (A.13) and (A.14), $\hat{L}$ and $\hat{U}$ are nonsingular, and

$$mnu \le \frac{1}{64}, \quad n(n+1)u \le \frac{1}{64}.$$

From (A.5), $\hat{Q}_1 = (A + E_5)\hat{R}_1^{-1}$, and

$$\begin{aligned}\hat{Q}_1^T\hat{Q}_1 &= \hat{R}^{-T}(A + E_5)^T(A + E_5)\hat{R}^{-1} \\ &= \hat{R}_1^{-T}A^TA\hat{R}_1^{-1} + \hat{R}_1^{-T}(E_5^TA + A^TE_5 + E_5^TE_5)\hat{R}_1^{-1}.\end{aligned} \tag{A.15}$$

Substituting (A.1) into (A.15) yields

$$\begin{aligned}&\hat{R}_1^{-T}A^TA\hat{R}_1^{-1}\\&=\hat{R}_1^{-T}(\hat{L}\hat{U}+E_1)^T(\hat{L}\hat{U}+E_1)\hat{R}_1^{-1}\\&=\hat{R}_1^{-T}\hat{U}^T\hat{L}^T\hat{L}\hat{U}\hat{R}_1^{-1}+\hat{R}_1^{-T}(E_1^T\hat{L}\hat{U}+\hat{U}^T\hat{L}^TE_1+E_1^TE_1)\hat{R}_1^{-1}.\end{aligned} \tag{A.16}$$

Moreover, substituting (A.2) and (A.3) into (A.16), we have

$$\hat{R}_1^{-T}\hat{U}^T\hat{L}^T\hat{L}\hat{U}\hat{R}_1^{-1}=\hat{R}_1^{-T}\hat{U}^T(\hat{S}^T\hat{S}-E_2-E_3)\hat{U}\hat{R}_1^{-1}. \tag{A.17}$$

From (A.4), we obtain

$$\begin{aligned}\hat{R}_1^{-T}\hat{U}^T\hat{S}^T\hat{S}\,\hat{U}\hat{R}_1^{-1}&=\hat{R}_1^{-T}(\hat{R}_1-E_4)^T(\hat{R}_1-E_4)\hat{R}_1^{-1}\\&=I-\hat{R}_1^{-T}E_4^T-E_4\hat{R}_1^{-1}+\hat{R}_1^{-T}E_4^TE_4\hat{R}_1^{-1}.\end{aligned} \tag{A.18}$$

Next, we introduce some lemmas.

Lemma 9. *The matrix $\hat{S}$ in Algorithm 3 satisfies*

$$\|\hat{S}^{-1}\|_2^2\le 1.02\sigma_{\min}(\hat{L})^{-2}. \tag{A.19}$$

Proof. From (A.2), (A.3), (A.9), and (A.10),

$$\hat{S}^T\hat{S}=\hat{B}+E_3=\hat{L}^T\hat{L}+E_2+E_3 \tag{A.20}$$

and

$$\begin{aligned}\sigma_{\min}(\hat{S})^2=\sigma_{\min}(\hat{L}^T\hat{L}+E_2+E_3)&\ge\sigma_{\min}(\hat{L})^2-\|E_2\|_2-\|E_3\|_2\\&\ge\sigma_{\min}(\hat{L})^2-mnu\|\hat{L}\|_2^2-n(n+1)u\|\hat{S}\|_2^2.\end{aligned} \tag{A.21}$$

From (A.9), (A.10), and (A.20),

$$\|\hat{S}\|_2^2-n(n+1)u\|\hat{S}\|_2^2\le\|\hat{L}\|_2^2+mnu\|\hat{L}\|_2^2,$$

and

$$\|\hat{S}\|_2^2\le\frac{1+mnu}{1-n(n+1)u}\|\hat{L}\|_2^2. \tag{A.22}$$

Hence, from (A.21) and (A.22),

$$\begin{aligned}\sigma_{\min}(\hat{S})^2&\ge\sigma_{\min}(\hat{L})^2-mnu\|\hat{L}\|_2^2-n(n+1)u\frac{1+mnu}{1-n(n+1)u}\|\hat{L}\|_2^2\\&\ge\sigma_{\min}(\hat{L})^2-(mn+n(n+1))u\frac{1+mnu}{1-n(n+1)u}\|\hat{L}\|_2^2.\end{aligned}$$

From the assumption (A.13),

$$64(mn+n(n+1))u\|\hat{L}\|_2^2\le\sigma_{\min}(\hat{L})^2.$$

Then,

$$\sigma_{\min}(\hat{S})^2\ge\sigma_{\min}(\hat{L})^2-\frac{\sigma_{\min}(\hat{L})^2}{64}\cdot\frac{1+1/64}{1-1/64}\ge0.983\sigma_{\min}(\hat{L})^2. \tag{A.23}$$

□

Lemma 10. *The matrix $\hat{R}_1$ in Algorithm 3 satisfies*

$$\|\hat{R}_1^{-1}\|_2\le1.03\sigma_{\min}(\hat{L})^{-1}\sigma_{\min}(\hat{U})^{-1}. \tag{A.24}$$

Proof. From (A.22),

$$\|\hat{S}\|_2^2\le\frac{1+mnu}{1-n(n+1)u}\|\hat{L}\|_2^2\le\frac{65}{63}\|\hat{L}\|_2^2$$

and from (A.4) and (A.11),

$$\begin{aligned}\sigma_{\min}(\hat{R}_1)=\sigma_{\min}(\hat{S}\,\hat{U}+E_4)&\ge\sigma_{\min}(\hat{S}\,\hat{U})-\|E_4\|_2\\&\ge\sigma_{\min}(\hat{S})\sigma_{\min}(\hat{U})-n^2u\|\hat{S}\|_2\|\hat{U}\|_2.\end{aligned}$$

From these, (A.14), and (A.23),

$$\begin{aligned}\sigma_{\min}(\hat{R}_1)&\ge\sigma_{\min}(\hat{S})\sigma_{\min}(\hat{U})-n^2u\|\hat{S}\|_2\|\hat{U}\|_2\\&\ge\sqrt{0.983}\sigma_{\min}(\hat{L})\sigma_{\min}(\hat{U})-\sqrt{\frac{65}{63}}n^2u\|\hat{L}\|_2\|\hat{U}\|_2\\&\ge0.99\sigma_{\min}(\hat{L})\sigma_{\min}(\hat{U})-\sqrt{\frac{65}{63}}\frac{\sigma_{\min}(\hat{L})\sigma_{\min}(\hat{U})}{64}\\&\ge0.974\sigma_{\min}(\hat{L})\sigma_{\min}(\hat{U}).\end{aligned}$$

□

Lemma 11. *The matrix $\hat{R}_1$ in Algorithm 3 satisfies*

$$\kappa_2(\hat{R}_1)\le1.03(1+n^2u)\sqrt{\frac{1+mnu}{1-n(n+1)u}}\kappa_2(\hat{L})\kappa_2(\hat{U}). \tag{A.25}$$

Proof. From (A.4) and (A.11), and (A.22),

$$\|\hat{R}_1\|_2\le(1+n^2u)\|\hat{S}\|_2\|\hat{U}\|_2$$

and

$$\|\hat{S}\|_2^2\le\frac{1+mnu}{1-n(n+1)u}\|\hat{L}\|_2^2. \tag{A.26}$$

Therefore,

$$\|\hat{R}_1\|_2\le(1+n^2u)\sqrt{\frac{1+mnu}{1-n(n+1)u}}\|\hat{L}\|_2\|\hat{U}\|_2.$$

Combining this and Lemma 10 proves the lemma. □

Lemma 12. *The matrices $\hat{L},\hat{U},\hat{R}_1$ in Algorithm 3 satisfy*

$$\|\hat{L}\hat{U}\hat{R}_1^{-1}\|_2^2\le1.06.$$

Proof. From (A.22) and Lemma 10,

$$\begin{aligned}\|\hat{S}\|_2\|\hat{U}\|_2\|\hat{R}_1^{-1}\|_2&\le\sqrt{\frac{1+mnu}{1-n(n+1)u}}\|\hat{L}\|_2\|\hat{U}\|_2\|\hat{R}_1^{-1}\|_2\\&\le1.03\sqrt{\frac{65}{63}}\kappa_2(\hat{L})\kappa_2(\hat{U}).\end{aligned}$$

From this, (A.4), (A.11), and Lemma 9,

$$\begin{aligned}\|\hat{U}\hat{R}_1^{-1}\|_2&\le\|\hat{S}^{-1}\|_2(1+\|E_4\|_2\|\hat{R}_1^{-1}\|_2)\\&\le\|\hat{S}^{-1}\|_2(1+n^2u\|\hat{S}\|_2\|\hat{U}\|_2\|\hat{R}_1^{-1}\|_2)\\&\le1.01\sigma_{\min}(\hat{L})^{-1}\left(1+1.03\sqrt{\frac{65}{63}}n^2u\kappa_2(\hat{L})\kappa_2(\hat{U})\right).\end{aligned}$$

From the assumption (A.14),

$$\|\hat{U}\hat{R}_1^{-1}\|_2 \le 1.01\sigma_{\min}(\hat{L})^{-1}\left(1+\frac{1.03}{64}\sqrt{\frac{65}{63}}\right) \le 1.03\sigma_{\min}(\hat{L})^{-1}. \tag{A.27}$$

Moreover, from (A.4), (A.17), and (A.18),

$$\begin{aligned}\|\hat{L}\hat{U}\hat{R}_1^{-1}\|_2^2 &= \|\hat{R}_1^{-T}\hat{U}^T\hat{L}^T\hat{L}\hat{U}\hat{R}_1^{-1}\|_2 \\ &\le \|\hat{R}_1^{-T}\hat{U}^T\hat{S}^T\hat{S}\hat{U}\hat{R}_1^{-1}\|_2 + \|\hat{R}_1^{-T}\hat{U}^T(E_2+E_3)\hat{U}\hat{R}_1^{-1}\|_2 \\ &\le \|(\hat{R}_1 - E_4)\hat{R}_1^{-1}\|_2^2 + \|\hat{U}\hat{R}_1^{-1}\|_2^2(\|E_2\|_2 + \|E_3\|_2) \\ &\le (1+\|E_4\|_2\|\hat{R}_1^{-1}\|_2)^2 + \|\hat{U}\hat{R}_1^{-1}\|_2^2(\|E_2\|_2+\|E_3\|_2).\end{aligned}$$

From (A.11), (A.22), and Lemma 10,

$$\begin{aligned}\|E_4\|_2\|\hat{R}_1^{-1}\|_2 &\le 1.03n^2u\|\hat{S}\|_2\|\hat{U}\|_2\sigma_{\min}(\hat{L})^{-1}\sigma_{\min}(\hat{U})^{-1} \\ &\le 1.03n^2u\frac{1+mnu}{1-n(n+1)u}\kappa_2(\hat{L})\kappa_2(\hat{U}),\end{aligned}$$

and, form (A.9), (A.10), and (A.22),

$$\begin{aligned}\|E_2\|_2 + \|E_3\|_2 &\le mnu\|\hat{L}\|_2^2 + n(n+1)u\|\hat{S}\|_2^2 \\ &\le \frac{1+mnu}{1-n(n+1)u}(mn+n(n+1))u\|\hat{L}\|_2^2\end{aligned}$$

Therefore, from these and (A.27),

$$\begin{aligned}\|\hat{L}\hat{U}\hat{R}_1^{-1}\|_2^2 \le &\left(1+\frac{1.03\cdot 65}{63}n^2u\kappa_2(\hat{L})\kappa_2(\hat{U})\right)^2 \\ &+\frac{1.07\cdot 65}{63}(mn+n(n+1))u\kappa_2(\hat{L})^2.\end{aligned}$$

Then, from (A.13) and (A.14),

$$\|\hat{L}\hat{U}\hat{R}_1^{-1}\|_2^2 \le \left(1+\frac{1.03\cdot 65}{64\cdot 63}\right)^2 + \frac{1.07\cdot 65}{63\cdot 64} \le 1.06.$$

□

Lemma 13. *The matrix $\hat{Q}_1$ computed by Algorithm 3 satisfies*

$$\|\hat{Q}_1\|_2 \le 1.112 =: \beta. \tag{A.28}$$

Proof. From (A.5) and (A.12),

$$\begin{aligned}\|\hat{Q}_1\|_2 &\le \|A\hat{R}_1^{-1}\|_2 + \|E_5\hat{R}_1^{-1}\|_2 \\ &\le \|A\hat{R}_1^{-1}\|_2 + n^2u\|\hat{Q}_1\|_2\kappa_2(\hat{R}_1),\end{aligned}$$

then

$$(1-n^2u\kappa_2(\hat{R}_1))\|\hat{Q}_1\|_2 \le \|A\hat{R}_1^{-1}\|_2. \tag{A.29}$$

From (A.8) and (A.16),

$$\begin{aligned}\|A\hat{R}_1^{-1}\|_2^2 &= \|\hat{R}_1^{-T}A^TA\hat{R}_1^{-1}\|_2 \\ &\le \|\hat{L}\hat{U}\hat{R}_1^{-1}\|_2^2 + 2\|E_1\|_2\|R_1^{-1}\|_2\|\hat{L}\hat{U}\hat{R}_1^{-1}\|_2 + \|E_1\|_2^2\|\hat{R}_1^{-1}\|_2^2 \\ &\le \|\hat{L}\hat{U}\hat{R}_1^{-1}\|_2^2 + 2n^2u\|\hat{L}\|_2\|\hat{U}\|_2\|\hat{R}_1^{-1}\|_2\|\hat{L}\hat{U}\hat{R}_1^{-1}\|_2 \\ &\quad + n^4u^2\|\hat{L}\|_2^2\|\hat{U}\|_2^2\|\hat{R}_1^{-1}\|_2^2.\end{aligned}$$

Here, from Lemmas 10 and 12,

$$\|\hat{L}\hat{U}\hat{R}_1^{-1}\|_2^2 \le 1.06, \quad \|\hat{R}_1^{-1}\|_2 \le 1.03\sigma_{\min}(\hat{L})^{-1}\sigma_{\min}(\hat{U})^{-1}. \tag{A.30}$$

Hence,

$$\begin{aligned}\|A\hat{R}_1^{-1}\|_2^2 \le &1.06 + 2\cdot 1.03\sqrt{1.06}n^2u\kappa_2(\hat{L})\kappa_2(\hat{U}) \\ &+ (1.03\cdot n^2u\kappa_2(\hat{L})\kappa_2(\hat{U}))^2.\end{aligned}$$

From the assumption (A.14),

$$\|A\hat{R}_1^{-1}\|_2^2 \le 1.06 + \frac{2\cdot 1.03\sqrt{1.06}}{64} + \left(\frac{1.03}{64}\right)^2 \le 1.094. \tag{A.31}$$

Also, from (A.14) and (A.25),

$$\begin{aligned}n^2u\kappa_2(\hat{R}_1) &\le 1.03(1+n^2u)\sqrt{\frac{1+mnu}{1-n(n+1)u}}n^2u\kappa_2(\hat{L})\kappa_2(\hat{U}) \\ &\le 1.03\cdot\frac{65}{64}\cdot\sqrt{\frac{65}{63}}\cdot\frac{1}{64}.\end{aligned}$$

From this, (A.29), and (A.31), we obtain

$$\|\hat{Q}_1\|_2 \le \frac{\|A\hat{R}_1^{-1}\|_2}{1-n^2u\kappa_2(\hat{R}_1)} \le 1.112. \tag{A.32}$$

□

Appendix A.2. Proof of Theorem 1

We estimate $\|\hat{Q}_1^T\hat{Q}_1 - I\|_2$, where $\hat{Q}_1$ is computed by Algorithm 3. From (A.15), (A.16), and (A.17), let

$$\begin{aligned}\delta_1 &:= \|\hat{R}_1^{-T}(E_5^TA + A^TE_5 + E_5^TE_5)\hat{R}_1^{-1}\|_2, \\ \delta_2 &:= \|\hat{R}_1^{-T}(-\hat{U}^T(E_2+E_3)\hat{U} + \hat{U}^T\hat{L}^TE_1 + E_1^T\hat{L}\hat{U} + E_1^TE_1)\hat{R}_1^{-1}\|_2, \\ \delta_3 &:= \|\hat{R}_1^{-T}(-E_4^T\hat{R}_1 - \hat{R}_1^TE_4 + E_4^TE_4)\hat{R}_1^{-1}\|_2.\end{aligned}$$

Then,

$$\|\hat{Q}_1^T\hat{Q}_1 - I\| \le \delta_1 + \delta_2 + \delta_3.$$

We first estimate δ_1. From (A.12) and (A.28),

$$\|E_5\|_2 \le n^2u\|\hat{Q}_1\|_2\|\hat{R}_1\|_2 \le \beta n^2u\|\hat{R}_1\|_2.$$

From this and (A.31),

$$\begin{aligned}\delta_1 &\le 2\|\hat{R}_1^{-1}\|_2\|E_5\|_2\|A\hat{R}_1^{-1}\|_2 + \|\hat{R}_1^{-1}\|_2^2\|E_5\|_2^2 \\ &\le 2\sqrt{1.094}\beta n^2u\kappa_2(\hat{R}_1) + \beta^2n^4u^2\kappa_2(\hat{R}_1)^2.\end{aligned} \tag{A.33}$$

Substituting (A.14) and (A.25) into (A.33),

$$\begin{aligned}\delta_1 &\le 2.06\sqrt{1.094}\beta n^2u(1+n^2u)\sqrt{\frac{1+mnu}{1-n(n+1)u}}\kappa_2(\hat{L})\kappa_2(\hat{U}) \\ &\quad + \beta^2n^4u^2\left(1.03(1+n^2u)\sqrt{\frac{1+mnu}{1-n(n+1)u}}\kappa_2(\hat{L})\kappa_2(\hat{U})\right)^2 \\ &= \frac{2.06}{64}\sqrt{1.094}\beta(1+n^2u)\sqrt{\frac{1+mnu}{1-n(n+1)u}}\delta_{LU} \\ &\quad + \frac{\beta^2}{64^2}1.03^2(1+n^2u)^2\frac{1+mnu}{1-n(n+1)u}\delta_{LU}^2 \\ &\le 0.0393\delta_{LU} + 0.00035\delta_{LU}^2 \le 0.0397\delta_{LU}.\end{aligned} \tag{A.34}$$

We next estimate δ_2. From (A.8), (A.9), (A.10), (A.22), (A.27), and Lemmas 10 and 12,

$$\begin{aligned}
\delta_2 &\le \|\hat{U}\hat{R}_1^{-1}\|_2^2(\|E_2\|_2 + \|E_3\|_2) \\
&\quad + 2\|E_1\|_2\|\hat{L}\hat{U}\hat{R}_1^{-1}\|_2\|\hat{R}_1^{-1}\|_2 + \|\hat{R}_1^{-1}\|_2^2\|E_1\|_2^2 \\
&\le 1.03^2\sigma_{\min}(\hat{L})^{-2}(mn\|\hat{L}\|_2^2 + n(n+1)\|\hat{S}\|_2^2)u \\
&\quad + 2.06n^2u\|\hat{L}\|_2\|\hat{U}\|_2 \cdot 1.03\sigma_{\min}(\hat{L})^{-1}\sigma_{\min}(\hat{U})^{-1} \\
&\quad + 1.03^2\sigma_{\min}(\hat{L})^{-2}\sigma_{\min}(\hat{U})^{-2}n^4u^2\|\hat{L}\|_2^2\|\hat{U}\|_2^2 \\
&\le 1.03^2\sigma_{\min}(\hat{L})^{-2}(mn + \frac{(1+mnu)n(n+1)}{1-n(n+1)u})u\|\hat{L}\|_2^2 \\
&\quad + 2.06n^2u\|\hat{L}\|_2\|\hat{U}\|_2 \cdot 1.03\sigma_{\min}(\hat{L})^{-1}\sigma_{\min}(\hat{U})^{-1} \\
&\quad + 1.03^2\sigma_{\min}(\hat{L})^{-2}\sigma_{\min}(\hat{U})^{-2}n^4u^2\|\hat{L}\|_2^2\|\hat{U}\|_2^2 \\
&= 1.03^2(mn + \frac{(1+mnu)n(n+1)}{1-n(n+1)u})u\kappa_2(\hat{L})^2 \\
&\quad + 2.06n^2u \cdot 1.03\kappa_2(\hat{L})\kappa_2(\hat{U}) + 1.03^2n^4u^2\kappa_2(\hat{L})^2\kappa_2(\hat{U})^2 \\
&\le \frac{1.03^2}{64}\frac{1+mnu}{1-n(n+1)u}\delta_L^2 + \frac{2.06\cdot 1.03}{64}\delta_{LU} + \frac{1.03^2}{64^2}\delta_{LU}^2 \\
&\le 0.0172\delta_L^2 + 0.0333\delta_{LU} + 0.00026\delta_{LU}^2 \\
&\le 0.0172\delta_L^2 + 0.034\delta_{LU}. \qquad \text{(A.35)}
\end{aligned}$$

We finally estimate δ_3. From (A.11), (A.22), and Lemma 10,

$$\begin{aligned}
\delta_3 &\le 2\|\hat{R}_1^{-1}\|_2\|E_4\|_2 + \|\hat{R}_1^{-1}\|_2^2\|E_4\|_2^2 \\
&\le 2.06\sigma_{\min}(\hat{L})^{-1}\sigma_{\min}(\hat{U})^{-1}n^2u\|\hat{S}\|_2\|\hat{U}\|_2 \\
&\quad + 1.03^2\sigma_{\min}(\hat{L})^{-2}\sigma_{\min}(\hat{U})^{-2}n^4u^2\|\hat{S}\|_2^2\|\hat{U}\|_2^2 \\
&\le 2.06\sigma_{\min}(\hat{L})^{-1}\sigma_{\min}(\hat{U})^{-1}n^2u\sqrt{\frac{1+mnu}{1-n(n+1)u}}\|\hat{L}\|_2\|\hat{U}\|_2 \\
&\quad + 1.03^2\sigma_{\min}(\hat{L})^{-2}\sigma_{\min}(\hat{U})^{-2}n^4u^2\frac{1+mnu}{1-n(n+1)u}\|\hat{L}\|_2^2\|\hat{U}\|_2^2 \\
&= \frac{2.06}{64}\sqrt{\frac{1+mnu}{1-n(n+1)u}}\delta_{LU} + \frac{1.03^2}{64^2}\frac{1+mnu}{1-n(n+1)u}\delta_{LU}^2 \\
&\le 0.0328\delta_{LU} + 0.000268\delta_{LU}^2 \le 0.034\delta_{LU}. \qquad \text{(A.36)}
\end{aligned}$$

Thus, combining (A.34), (A.35), and (A.36),

$$\begin{aligned}
\|\hat{Q}_1^T\hat{Q}_1 - I\|_2 &\le \delta_1 + \delta_2 + \delta_3 \\
&\le 0.0397\delta_{LU} + 0.0172\delta_L^2 + 0.034\delta_{LU} + 0.034\delta_{LU} \\
&\le 0.1077\delta_{LU} + 0.0172\delta_L^2 \le 0.1249\max(\delta_{LU}, \delta_L^2) \\
&\le \frac{1}{8}\max(\delta_{LU}, \delta_L^2),
\end{aligned}$$

which proves Theorem 1.

Appendix A.3. Proof of Corollary 2

We estimate $\|\hat{Q}_2^T\hat{Q}_2 - I\|_2$, where $\hat{Q}_2$ is computed by Algorithm 5. In a similar way to a rounding error analysis of CholeskyQR2 [8], an upper bound of $\|\hat{Q}_2^T\hat{Q}_2 - I\|_2$ can be obtained.

From (A.13), (A.14), and (7),

$$\|\hat{Q}_1^T\hat{Q}_1 - I\|_2 \le \frac{1}{8}.$$

Then,

$$\sqrt{1-\frac{1}{8}} \le \sigma_{\min}(\hat{Q}_1), \quad \sigma_{\max}(\hat{Q}_1) \le \sqrt{1+\frac{1}{8}}$$

and

$$\kappa_2(\hat{Q}_1) = \frac{\sigma_{\max}(\hat{Q}_1)}{\sigma_{\min}(\hat{Q}_1)} \le 1.134. \qquad \text{(A.37)}$$

From the assumption

$$\alpha := 8\kappa_2(\hat{Q}_1)\sqrt{mnu + n(n+1)u} \le 1,$$

it holds from (2) that

$$\|\hat{Q}_2^T\hat{Q}_2 - I\|_2 \le \frac{5}{64}\alpha^2. \qquad \text{(A.38)}$$

From this and (A.37),

$$\begin{aligned}
\|\hat{Q}_2^T\hat{Q}_2 - I\|_2 &\le \frac{5}{64}\left(8\cdot 1.134\sqrt{(mn+n(n+1))u}\right)^2 \\
&\le 6.5(mn+n(n+1))u,
\end{aligned}$$

which proves Corollary 2.

Appendix A.4. Proof of Theorem 3

From Lemmas 5 and 8, $\hat{Q}_1$, $\hat{R}_1$, $\hat{Q}_2$, $\hat{R}_2$, and $\tilde{R}$ in Algorithm 5 (through Algorithms 1 and 3) satisfy

$$\hat{Q}_2\tilde{R} - \hat{Q}_1 = E_6, \quad |E_6| \le nu|\hat{Q}_2||\tilde{R}|, \qquad \text{(A.39)}$$

$$\tilde{R}\hat{R}_1 - \hat{R}_2 = E_7, \quad |E_7| \le nu|\tilde{R}||\hat{R}_1|, \qquad \text{(A.40)}$$

$$\hat{Q}_1\hat{R}_1 - A = E_8, \quad |E_8| \le nu|\hat{Q}_1||\hat{R}_1|. \qquad \text{(A.41)}$$

Suppose that $\hat{R}_1$ is non-singular, then $\tilde{R}$ and $\hat{Q}_1$ satisfy

$$\tilde{R} = (\hat{R}_2 + E_7)\hat{R}_1^{-1}, \quad \hat{Q}_1 = (A + E_8)\hat{R}_1^{-1} \qquad \text{(A.42)}$$

from (A.40) and (A.41). To substitute (A.42) into (A.39), the residual can be estimated as follows.

$$\begin{aligned}
\hat{Q}_2(\hat{R}_2 + E_7)\hat{R}_1^{-1} - (A + E_8)\hat{R}_1^{-1} &= E_6 (= \hat{Q}_2\tilde{R} - \hat{Q}_1), \\
\hat{Q}_2(\hat{R}_2 + E_7) - A - E_8 &= E_6\hat{R}_1, \\
\hat{Q}_2\hat{R}_2 - A &= E_6\hat{R}_1 - \hat{Q}_2E_7 + E_8.
\end{aligned}$$

With this, the norm of residual is bounded by

$$\begin{aligned}
\|\hat{Q}_2\hat{R}_2 - A\|_2 &\le \|E_6\|_2\|\hat{R}_1\|_2 + \|\hat{Q}_2\|_2\|E_7\| + \|E_8\|_2 \\
&\le n^2u(2\|\hat{Q}_2\|_2\|\tilde{R}\|_2 + \|\hat{Q}_1\|_2)\|\hat{R}_1\|_2 \qquad \text{(A.43)}
\end{aligned}$$

from (A.6). Also, $\|\tilde{R}\|_2^2$ satisfies

$$\|\tilde{R}\|_2^2 \le \frac{1+mnu}{1-n(n+1)u}\|\hat{Q}_1\|_2 \le \frac{65}{63}\|\hat{Q}_1\|_2 \qquad \text{(A.44)}$$

as the same of proof of (A.26), since $\hat{S}$ and $\tilde{R}$ are Cholesky factors of L^TL and $\hat{Q}_1^T\hat{Q}_1$ respectively. From (A.28) and (A.38),

$$\|\hat{Q}_1\|_2 \le 1.112, \quad \|\hat{Q}_2\|_2 \le \sqrt{1+\frac{5}{64}} \le 1.0384.$$

Therefore, from (A.43) and (A.44),

$$||\hat{Q}_2\hat{R}_2 - A||_2 \leq 3.495n^2u||\hat{R}_1||_2. \tag{A.45}$$

Next, we analyze upper bound of $||\hat{R}_1||_2$. From Theorem 1,

$$\sigma_{\min}(\hat{Q}_1) \geq \sqrt{1-\frac{1}{8}} \geq 0.875, \sigma_{\max}(\hat{Q}_1) \leq \sqrt{1+\frac{1}{8}} \leq 1.125.$$

From this and (A.41), we obtain

$$\begin{aligned}||\hat{Q}_1\hat{R}_1 - E_8||_2 &\geq ||\hat{Q}_1\hat{R}_1||_2 - n^2u||\hat{Q}_1||_2||\hat{R}_1||_2 \\ &\geq \sigma_{\min}(\hat{Q}_1)||\hat{R}_1||_2 - n^2u||\hat{Q}_1||_2||\hat{R}_1||_2\end{aligned}$$

and

$$\begin{aligned}||\hat{R}_1||_2 &\leq \frac{||A||_2}{\sigma_{\min}(\hat{Q}_1) - n^2u||\hat{Q}_1||_2} \\ &\leq \frac{||A||_2}{0.875 - 1.125n^2u} \leq 1.17||A||_2. \end{aligned}\tag{A.46}$$

To substitute (A.46) into (A.45),

$$\begin{aligned}||\hat{Q}_2\hat{R}_2 - A||_2 &\leq 3.495 \cdot 1.17n^2u||A||_2 \\ &\leq 4.09n^2u||A||_2.\end{aligned}$$

Appendix A.5. Proof of Theorem 4

From (A.1), (A.4), and (A.6),

$$\hat{L}\hat{U} - PA = E_1, \quad ||E_1|| \leq n^2u||\hat{L}||_2||\hat{U}||_2, \tag{A.47}$$

$$\hat{R}_1 - \hat{S}\,\hat{U} = E_4, \quad ||E_4|| \leq n^2u||\hat{S}\,||_2||\hat{U}||_2, \tag{A.48}$$

and from Lemma 8,

$$\hat{Q}_1\hat{S} - P^T\hat{L} = E_9, \quad ||E_9||_2 \leq n^2u||\hat{Q}_1||_2||\hat{S}\,||_2. \tag{A.49}$$

From (A.22), (A.48), and (A.49), we obtain

$$||E_4|| \leq n^2\frac{1+mnu}{1-n(n+1)u}u||\hat{L}||_2||\hat{U}||_2 \leq \frac{65}{63}n^2u||\hat{L}||_2||\hat{U}||_2, \tag{A.50}$$

$$||E_9|| \leq \frac{65}{63}n^2u||\hat{Q}_1||_2||\hat{L}||_2. \tag{A.51}$$

From (A.49),

$$\hat{Q}_1\hat{S}\,\hat{U} - P^T\hat{L}\hat{U} = E_9\hat{U},$$

and from (A.47) and (A.48),

$$\begin{aligned}\hat{Q}_1(\hat{R}_1 - E_4) - A - P^TE_1 &= E_9\hat{U}, \\ \hat{Q}_1\hat{R}_1 - A &= \hat{Q}_1E_4 + P^TE_1 + E_9\hat{U}.\end{aligned}$$

From this, (A.47), (A.50), and (A.51),

$$\begin{aligned}||\hat{Q}_1\hat{R}_1 - A||_2 &\leq ||\hat{Q}_1||_2||E_4||_2 + ||E_1||_2 + ||E_9||_2||\hat{U}||_2 \\ &\leq n^2u(2\frac{65}{63}||\hat{Q}_1||_2||\hat{L}||_2||\hat{U}||_2 + ||\hat{L}||_2||\hat{U}||_2),\end{aligned}$$

and, from (2),

$$||\hat{Q}_1\hat{R}_1 - A||_2 \leq (1 + 2\frac{65}{63}\sqrt{1+\frac{5}{64}})n^2u||\hat{L}||_2||\hat{U}||_2 \tag{A.52}$$

$$\leq 3.15n^2u||\hat{L}||_2||\hat{U}||_2. \tag{A.53}$$

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